

Smart environmental DNA sampling strategies: Spatial autocorrelation can guide efficient survey design

Running headline: Efficient eDNA survey design

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8 **Abstract**

9 1. Environmental DNA (eDNA) surveys are increasingly used to monitor marine biodiversity, yet
10 managers have little quantitative guidance on how many samples to collect. Because ship time is
11 costly, it matters where extra sampling stops improving the information used for decisions; and
12 because species differ in patchiness, no single sample size serves all.

13 2. We combined eDNA metabarcoding data for 12 fish species from 358 stations and five depth
14 strata along the U.S. West Coast with 200 simulated spatial fields spanning autocorrelation length-
15 scales (ρ) of 25–1000 km. Bayesian Gaussian process (GP) models estimated species-specific
16 spatial structure, and datasets were progressively thinned to quantify how reduced effort degrades
17 prediction and parameter recovery.

18 3. As expected, predictions improved non-linearly with increasing sampling density and with
19 increasing degrees of autocorrelation (ρ).

20 4. Marginal gains became negligible at 4–5 samples per 10,000 km² per depth layer for smoothly
21 distributed species ($\rho > 500$ km), whereas patchily distributed species ($\rho < 50$ km) required 8–11
22 samples per 10,000 km² — over twice the effort for equal precision. More generally, these
23 thresholds reflect sample spacing relative to the species' spatial autocorrelation length-scale,
24 providing a scale-independent basis for translating sampling requirements across systems.

25 5. The inferential objective mattered as much as the effort: depth-specific means, which underpin
26 abundance indices, were recovered reliably under sparse sampling, whereas estimating ρ required
27 denser designs spanning a wide range of distances, stabilising above ~6 samples per 10,000 km².

28 6. *Synthesis and applications.* Optimal eDNA sampling effort depends on the patchiness of the
29 target taxon and on the purpose of the survey. Our error model can be inverted to convert a required
30 precision and a prior estimate of spatial autocorrelation into a minimum sampling density, letting

survey designers allocate ship time before a cruise. Across the California Current, 4–9 samples per 10,000 km² per depth layer spanned the point of diminishing returns, modestly above the density of existing regional survey grids (3.4–3.6 samples per 10,000 km²). Programmes producing abundance indices for stock assessment can operate near current effort; those mapping fine-scale distributions need denser designs including clusters of closely spaced stations.

Keywords: biodiversity monitoring, environmental DNA, fisheries management, Gaussian process, metabarcoding, sampling design, spatial autocorrelation

Introduction

Biodiversity monitoring programmes underpin decisions about where to fish, what to protect and how to report progress against national and international conservation targets, yet the budgets available to them are effectively fixed. How many samples to collect, and where, is therefore a management decision as much as a statistical one (Yoccoz et al., 2001). Survey design must balance the need for sufficient data for detecting trends and changes in population while also minimizing cost (i.e., maximizing sampling efficiency) (Stein & Ettema, 2003; Xu et al., 2015; Zhao et al., 2022).

The emergence of environmental DNA (eDNA) sampling has revolutionized monitoring capabilities, offering a non-invasive method to detect species presence and estimate relative abundance (Gold et al., 2021; Shelton et al., 2022; Taberlet et al., 2012). Recent synthesis work shows that eDNA methods have matured sufficiently for key uncertainties to be quantified and incorporated into existing fisheries stock-assessment frameworks, enabling the development of

cost-effective abundance indices that complement conventional surveys (Baetscher et al., 2025; Guri et al., 2024; Kirtane et al., 2021; Salter et al., 2026). Because agencies are now evaluating eDNA as an operational component of biodiversity assessment rather than as an exploratory tool, explicit and defensible guidance on how much to sample has become a prerequisite for uptake rather than a refinement of it.

While increased sampling intensity generally reduces variance in population estimates and improves the precision of spatial pattern detection in species distribution models (Schweiger et al., 2016), this relationship exhibits diminishing returns where beyond a certain threshold, additional sampling effort produces marginal gains in statistical precision that may not justify the associated costs and logistical demands (Perret et al., 2022; Zhao et al., 2022).

We conceptualize species spatial distributions as realizations of a Gaussian Process (GP), in which spatial dependence is characterized by a species-specific length-scale parameter (ρ) describing the distance over which observations remain correlated. We hypothesize that the sampling density required to characterize a species' distribution depends on this spatial length-scale. A species with a long length-scale varies gradually across space and can therefore be reconstructed from fewer samples, whereas a species with a short length-scale exhibits more rapid spatial variation and consequently requires denser sampling (Figure 1). Thus, knowledge of species-specific spatial autocorrelation can provide a quantitative basis for informing sampling effort (Fortin et al., 1989). In this study, we assess how species spatial autocorrelation influences the efficiency of eDNA metabarcoding marine sampling strategies. Using both simulated and empirical eDNA data for species that differ in their spatial distributions, we quantify how spatial autocorrelation and

sampling effort affect (i) the predictive accuracy of species distribution models and (ii) the accuracy of GP parameter estimation. Our objective is deliberately operational: to provide survey planners with guidance that can be applied before a cruise, based on a prior estimate of the spatial autocorrelation of target taxa and the level of precision required.

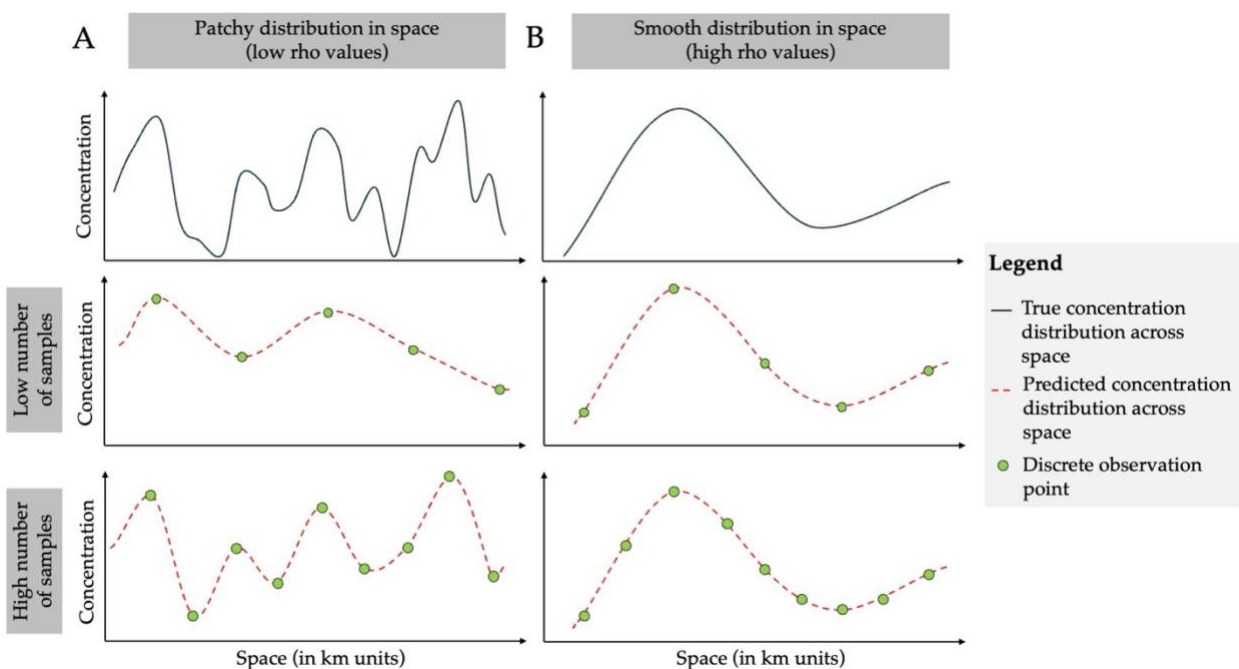


Figure 1. Conceptual illustration how sampling density affects reconstruction of spatial concentration patterns across different spatial structures. Panel A shows a patchy species distribution in space, whereas panel B shows a smooth distribution in space, along a one-dimensional spatial transect (x-axis; space in km units). In the top row of each panel, the solid black line represents the true underlying concentration distribution whereas the middle and bottom rows show reconstructions of concentration distribution (red dashed line) based on a low number of samples and a high number of samples, respectively. Green circles indicate discrete observation points.

Materials and methods

We evaluated how sampling effort influences the recovery of marine eDNA spatial patterns by integrating empirical observations with simulation-based analyses within a common GP framework. We first analyzed a real-world eDNA metabarcoding dataset collected across 358 stations and five depth strata along the U.S. West Coast to characterize species-specific spatial structure and depth-specific mean eDNA concentration patterns. We then used the range of parameter values informed by these empirical estimates to simulate new spatial eDNA fields across the same study domain, but at a much higher sampling resolution than was available in the field dataset. For both empirical and simulated data, we evaluated how reduced sampling effort affected spatial prediction accuracy and the recovery of GP parameters through systematic thinning analyses. By combining empirical and simulation-based evaluation, this framework allowed us to assess sampling efficiency beyond the limits of the empirical dataset alone (restricted to 358 observations) and thus to evaluate the sampling effort required for reliable inference across a broader range of effort.

Modelling framework: Gaussian processes

The GP approach is grounded in the assumption that species distributions and abundances change in a smooth and continuous manner across space (or other covariates), reflecting correlated responses to spatial, environmental, and biological covariates (Lewy & Kristensen, 2009). At the core, in simple terms, GPs posit that observations made at proximate locations or covariate values should be similar (Dormann et al., 2007). We assert that many species can be well described using relatively simple covariance models (Thorson et al., 2015). Specifically, we use an isotropic, squared exponential covariance model for the distribution of a species. Let $X = \{x_1, x_2, \dots, x_N\}$ be the

set of N observed spatial locations. Then species (eDNA) abundance or density Y across all sampled locations (X) is modeled as,

$$Y \sim MVN(\mu, \mathbf{K} + \sigma^2 \mathbf{I}), \quad 1$$

where μ is the global mean abundance, σ^2 is the non-spatial variance, $\mathbf{I}$ is an $n \times n$ identity matrix and $\mathbf{K}(X, X)$ is the covariance matrix whose $(i, j)^{\text{th}}$ entry is given by the squared exponential kernel function,

$$\mathbf{K}(X, X) = k(x_i, x_j) = \alpha^2 \exp\left(-\frac{(x_i - x_j)^2}{2\rho^2}\right) \quad 2$$

where α^2 is the amplitude (i.e., the marginal variance of the spatial process, controlling the overall magnitude of variation in abundance), ρ is the length-scale parameter (in $x_i - x_j$ units; hereafter spatial autocorrelation).

Empirical eDNA data

Environmental DNA samples were collected during the Joint U.S.-Canada Integrated Ecosystem and Pacific Hake Acoustic-Trawl Survey aboard the NOAA ship Bell M. Shimada in summer 2019 (de Blois 2020). Sampling spanned a broad latitudinal and longitudinal gradient along the U.S. West Coast, extending from 123.53° to 125.93° W (243–461 km UTM eastings; span = 218 km) and from 38.36° - 48.46° N (4249 - 5367 km UTM northings; range 1118 km), corresponding to a region from central California to northern Washington. At each site, seawater (10 L) was collected using Niskin bottles mounted on a CTD rosette system, targeting five depths (0, 50, 150, 300, and 500 m). From each bottle, 2.5 L of seawater was immediately filtered onboard through sterile 47 mm mixed cellulose ester filters (1 μm pore size) to capture suspended eDNA. Filters were preserved and DNA was extracted using a phase-lock phenol-chloroform protocol, following established methods (Ramón-Laca et al., 2021; Shelton et al., 2022).

Samples underwent PCR amplification using the MiFish U universal primer set (Miya et al., 2015), targeting a 170 bp fragment of the mitochondrial 12S rRNA gene and thereafter sequenced on an Illumina MiSeq platform (2x150 bp paired-end) at the University of Washington Genomics Core. Raw sequence reads were processed using the DADA2 pipeline (Callahan et al., 2016) to generate amplicon sequence variants (ASVs), which were taxonomically assigned using a custom reference database from NCBI GenBank and BOLD Systems. The same set of samples were used to estimate hake eDNA concentration using qPCR (see more Guri et al., 2025). Species-specific eDNA concentrations were estimated by jointly modeling ASV read counts, a mock community to correct for amplification bias, and qPCR measurements of Pacific hake to estimate absolute eDNA abundance for 12 fish species (Guri et al., 2024, 2025). The focal assemblage comprised a mixture of pelagic, mesopelagic, and demersal taxa characteristic of the California Current ecosystem, including *Clupea pallasii*, *Engraulis mordax*, *Leuroglossus stilbius*, *Microstomus pacificus*, *Sardinops sagax*, *Scomber japonicus*, *Sebastes entomelas*, *Stenobranchius leucopsarus*, *Tactostoma macropus*, *Tarletonbeania crenularis*, *Thaleichthys pacificus* and *Trachurus symmetricus*. The final empirical dataset included eDNA measurements (in log copies/L) from 358 spatially distributed sampling among the reported depth layers.

Estimation of species-specific spatial structure

After the acquisition of the empirical data, we obtained log eDNA concentration values (Y) paired with spatial coordinates (X) for each sampling location (x ; $N = 358$). We then used the GP model (Eq. 1–2) to infer the underlying spatial structure and learn key parameters governing the spatial field, including the depth-specific mean (μ_d), spatial autocorrelation (ρ), amplitude (α^2), and non-

spatial variance (σ^2). Additionally, we used the model to predict eDNA concentrations (Z) across the full spatial grid (Q) using Eq 3. We defined this grid (Q), as 18,916 discrete point estimates, representing the centroids of approximately 8×8 km spatial cells distributed across five depth layers. The number of points differed among depth layers (4,000, 4,000, 4,000, 3,622, and 3,294 points, respectively) because deeper layers covered a smaller spatial extent than shallower layers due to bathymetry.

For a given species, we compute the covariance matrix among observed locations, $\mathbf{K}(X, X')$, using the squared exponential kernel and the non-spatial variance term ($\sigma^2 \mathbf{I}$). Using the Cholesky decomposition we solved for $(\mathbf{K}(X, X') + \sigma^2 \mathbf{I})^{-1}(Y - \mu_d)$ and we combined it with the cross-covariance between observed and prediction points, $\mathbf{K}(X, Q)$, to yield the predictions at each given spatial point (Q):

$$Z = \mu_d + \mathbf{K}(X, Q)^\top (\mathbf{K}(X, X') + \sigma^2 \mathbf{I})^{-1} (Y - \mu_d) \quad 3$$

Estimating sampling efficiency on prediction accuracy

To evaluate how sampling effort (E) influences spatial reconstruction and prediction accuracy, we generated a series of reduced-effort observation datasets (Y_E coupled with X_E) by randomly thinning the observation set (Y) and its spatial covariate (X_E) to fixed sample sizes ($N = 330, 300, 260, 220$, and 150 ; Figure 2B). Each reduced-effort observation dataset (Y_E) together with its covariate (X_E) was refit with the same GP model (Eq 1-2) and produced effort-specific spatial predictions (S_E) on the full grid Q , by substituting S_E for Z , Y_E for Y and X_E for X in Eq. 3. This framework enables direct, like-for-like comparisons of reconstructed fields across sampling

intensities since the predicted field is the same (Q). Sampling effort (E) was quantified as sampling density:

$$E = \frac{N}{A \times H} \times 10000 \quad 4$$

where A is the spatial area of the study domain (200,000 km²), and H is the number of depth layers (d) included in this study ($H=5$), yielding E in units of samples per 10,000 km² per depth layer.

For each model prediction, we calculated the prediction error (ε_{1E}), defined as the root mean squared error between spatial fields predicted from the full dataset (Z) and those predicted from each reduced sampling scenario (S_E) as follows (Figure 2B):

$$\varepsilon_{1E} = \sqrt{\frac{1}{n} \sum_{i=1}^n (Z_i - S_{iE})^2} \quad 5$$

with n being the total number of Q grid points ($n = 18,916$) where the predictions (Z and S_E) are projected. We repeated this thinning-fitting-prediction process across 20 random realizations for each thinning intensity.

Simulated eDNA data

To extend inference beyond the sampling effort limitations ($N = 358$) of the empirical dataset, we conducted a series of simulations. We generated *in-silico* eDNA concentration data using the GP model (Eq. 1-2) across the same spatial domain and prediction grid (Q) used in the empirical analysis.

The simulations were designed to explore a wide range of spatial structures by varying the GP length-scale parameter (ρ) across ten levels: 25, 50, 75, 100, 200, 300, 400, 500, 800, and 1000 km (Appendix S1: Table S1). For each value of ρ we generated (arbitrarily) 20 independent replicate spatial fields, resulting in a total of 200 simulated datasets (10 ρ levels $\times$ 20 replicates; Appendix S1: Table S1). Similar to GP inference of the empirical data, within each simulation, ρ was held constant across depth layers, but each layer had a separate mean (μ_d) drawn randomly from a normal distribution with mean of 0 and a standard deviation of 3. We focused only on spatial autocorrelation across latitude and longitude, allowing the eDNA concentration at each depth to be independent. The amplitude (α^2) and the non-spatial variance (σ^2) in Eq. 1 and 2 were fixed at 4 and 3, respectively, to mirror parameter estimates in the empirical data. Both values were fixed based on the empirical estimates, where α^2 ranged from 2 to 4 across species and σ^2 was approximately 3 (Appendix S1: Table S1).

Given the large number of projected eDNA points (Q), fitting a GP model using all locations ($\approx$ 20,000 input values) was computationally prohibitive. We therefore drew a subset of 4,000 points from each of the 200 simulated datasets as the “complete” observation dataset (Y) and its covariate (X). Following a procedure identical to that used for the empirical data, we first fit the GP model (Eq. 1–2) to each complete simulated dataset (Y) and covariate (X) to infer its governing parameters (μ_d , ρ , α^2 , and σ^2) and thereafter predict eDNA concentrations (Z) across the full spatial grid (Q) using Eq. 3. This prediction (Z) served as the “true” spatial field against which all subsequent reduced-effort predictions were compared (Figure 2A).

To evaluate how sampling effort influences prediction accuracy on the simulated data, we generated a series of reduced-effort observation datasets (Y_E) and covariates (X_E) by randomly thinning each complete set (Y) and its covariate (X) to progressively smaller sample sizes ($N = 2000, 1000, 600, 358, 330, 300, 260, 220, \text{ and } 150$; Figure 2A). Following the same procedure used for the empirical data, we fit the GP model (Eq. 1-3) to estimate the governing parameters and generate an effort-specific spatial prediction (S_E) on the full spatial grid (Q). We then quantified the prediction error (ϵ_{2E}) for each scenario by calculating the root mean squared error between the “true” field (Z) and the reduced-effort field (S_E), as defined in Eq. 5 (Figure 2A). This allowed us to quantify how accurately different sampling efforts (E) reconstructed the underlying eDNA field.

Additionally, to evaluate the influence of sampling effort on GP parameter estimation, we compared the estimated parameters ($\rho, \mu_d, \sigma^2, \alpha^2$) from effort-specific datasets (Y_E) and covariates (X_E) to the true values used to generate the simulated fields. Differences between estimated and simulated parameter values were quantified using the root mean squared error (hereafter τ). This allowed us to quantify how reliably different sampling effort (E) recovered the underlying governing parameters of GP models.

Regression analysis of prediction error

To examine how sampling effort and spatial autocorrelation jointly shape prediction error, we fit a log-linear model to each dataset separately (empirical: ϵ_1 ; simulated: ϵ_2), isolating each factor's contribution and enabling comparison across datasets.

We modeled prediction error (ε) as a function of sampling density (E) and spatial autocorrelation (ρ), including their interaction using a linear model (*lm* function) in R as follows:

$$\log(\varepsilon) = \omega + \beta \log(E) + \theta \log(\rho) + \gamma \log(E)\log(\rho) \quad 6$$

where ω is the intercept, β slope of sampling density (E), θ is the slope of spatial autocorrelation (ρ) and γ is the coefficient on the interaction between the two latter factors.

Comparison of prediction accuracy between empirical and simulations

Due to the gap in sampling effort between field data and simulation datasets, prediction accuracy was benchmarked differently between the two datasets. The empirical benchmark (ε_1) was determined by the sampling effort of $N = 358$, while the simulation accuracy (ε_2) was benchmarked against a much larger set of sampling effort $N = 4,000$ samples (Figure 2).

To enable a like-for-like comparison between simulated and empirical results, we performed a secondary thinning analysis and prediction accuracy estimation (ε_3) in which the benchmark was redefined to match empirical sampling effort ($N = 358$; Figure 2A). Taking the $N=358$ subsample as the reference observation set and its prediction on Q as the baseline (Z), we compared the predictions (S_E) of the sampling-effort datasets (Y_E ; $E = 330, 300, 260, 220$, and 150) to both the simulated and empirical datasets, refit the GP model at each effort, and computed effort-specific (ε_3) prediction errors by comparing each S_E to the matched reference field Z (derived from $S_{E=358}$; Figure 2A). Furthermore, we fitted the prediction accuracy (ε_3) as a function of sampling effort (E) and spatial autocorrelation (ρ) using Eq. 6 (similarly to empirical data). We then compared the estimated model coefficients of prediction accuracy between the simulated and empirical to assess whether similar patterns were observed.

270

271 All the GP models (equation 1 and 2) were fitted in a Bayesian framework using Stan probabilistic

272 programming language (Stan Development Team, 2023)8/24/2026 4:32:00 AM. We used weakly

273 informative priors (Table 1) to allow the data to primarily inform the posterior distributions of the

274 parameters. For each GP model in Stan, we ran four parallel chains with 2,000 iterations each,

275 discarding the first 1,000 iterations as warm-up. Stan's warm-up phase both discards these initial

276 draws and adapts the sampler's step size and mass matrix, and so is not strictly equivalent to a

277 conventional burn-in. Convergence was assessed using $\hat{R}$ values, where values below 1.01

278 indicated satisfactory convergence and effective sample size (ESS) above 200. The model in Eq.

279 6 was fitted using linear models using $lm()$ function in R (R Core Team, 2024; Appendix S1:

280 Figure S5 for model fit diagnostics).

281

282 *Table 1. Gaussian Process model parameters and their prior distributions used in Bayesian*

283 *inference.*

Parameter	Prior	Range	
		Min	Max
α^2	$\mathcal{N}(4,1)$	0	$+\infty$
μ	$\mathcal{N}(0,5)$	-5	$+\infty$
$\log(\rho)$	$\mathcal{N}(0,1)$	-5	$+\infty$
σ^2	$\mathcal{N}(0,1)$	0	$+\infty$

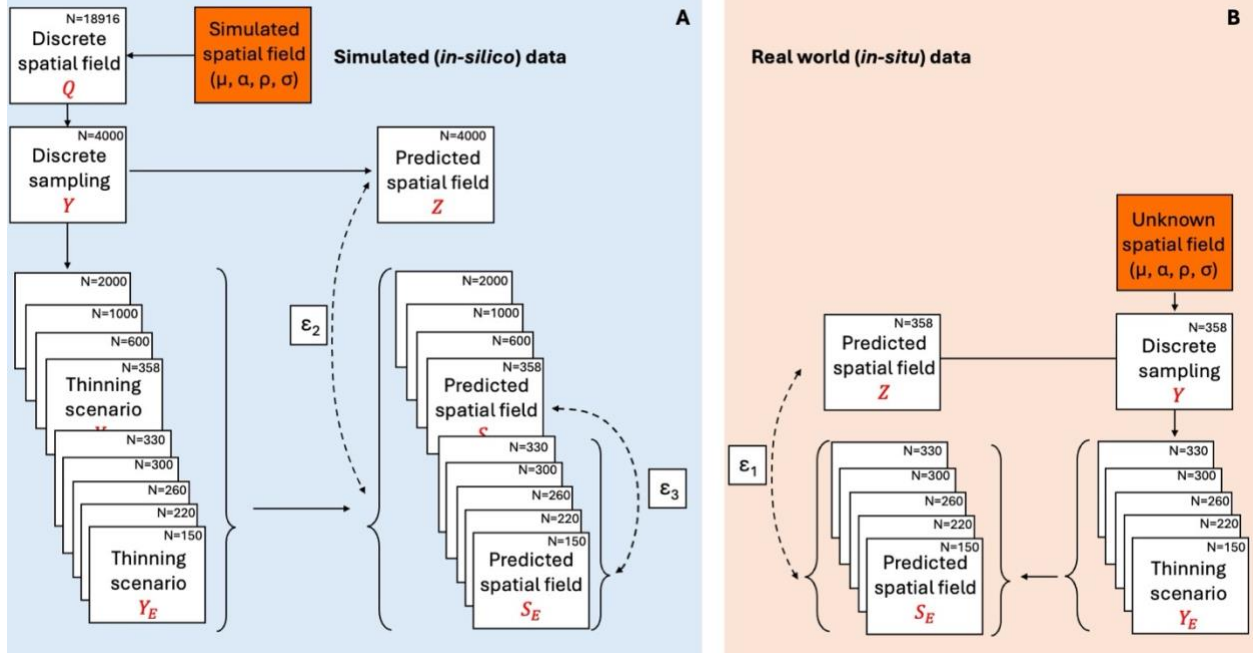


Figure 2. Workflow for evaluating sampling-density effects on spatial field reconstruction in simulated (in-silico) and empirical (in-situ) data. A spatial field is generated (panel A; in-silico data) from known parameters (μ, ρ, σ) and represented on a discrete grid (Q , $N=18,916$). A reference sampling set (Y , $N=4000$) is used to fit a GP model and generate a predicted field (Z), which serves as the reference reconstruction. The prediction error (ϵ_2) is defined as the difference between Z field and the progressively reduced using thinning designs (Y_E) reconstructed field. To enable a like-for-like comparison with the in-situ data, a secondary baseline using $N=358$ samples is defined to produce a reference prediction $S_{E=358}$, and prediction error (ϵ_3) is calculated as the difference between $S_{E=358}$ and reconstructions obtained from further thinned datasets (Y_E). Panel B (in-situ data) applies the same thinning and refitting to the observed dataset (Y , $N=358$); deviations of the thinned predictions from that baseline quantify ϵ_1 .

Results

Empirical data

299 Across species, depth-specific global mean eDNA concentrations generally declined with
300 increasing sampling depth (Figure 3). Species-specific length-scale estimates varied considerably
301 across species (Figure 3A), likely reflecting differences in habitat use and movement behavior.
302 Surface-dwelling species tended toward shorter spatial autocorrelation lengths and showed more
303 pronounced differences between surface and deep mean eDNA concentrations (μ ; Figure 3B),
304 though exceptions were evident. Deep-dwelling species broadly exhibited longer spatial
305 autocorrelations (Figure 3A) and relatively stable mean eDNA concentrations across depth strata,
306 although this pattern was not universal. *Tactostoma macropus*, for instance, showed a
307 comparatively short length-scale despite its deep-water affinity. Together, these patterns suggest
308 that surface-associated species may be more spatially heterogeneous, whereas deeper-dwelling
309 species may display increasingly homogeneous distributions.

310
311 Across species, estimated α^2 ranged from 2.3 to 4.2, whereas the GP non-spatial variance σ^2
312 spanned from 1.5 to 4.5. Surface-associated taxa generally exhibited slightly higher σ^2 than
313 bottom-associated taxa, indicating greater process variance of the inferred eDNA fields. This
314 pattern may partly reflect a mean–variance relationship, as surface taxa showed higher mean
315 eDNA concentrations or alternatively, differences in σ^2 are linked to the spatial autocorrelation
316 parameter ρ .

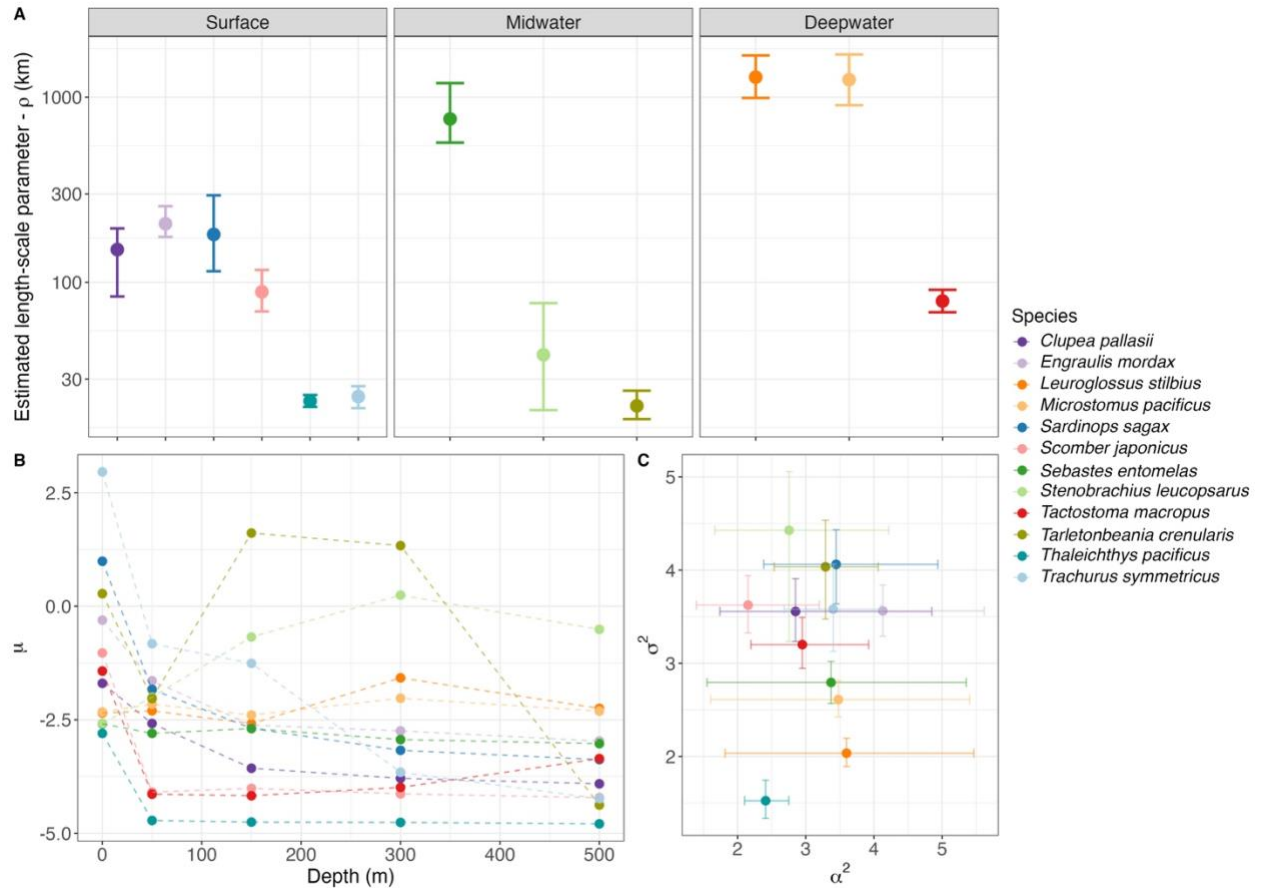


Figure 3. Species-specific Gaussian Process parameter estimates of empirical data. (A) Estimated spatial autocorrelation parameter (ρ in km) for each species (indicated in colors) across the three major depth strata defined from the literature (surface, midwater, and deepwater). Points indicate posterior mean estimates and bars denote 95% credible intervals. Depth-specific (0, 50, 100, 300, and 500 m) relationship between species spatial autocorrelation (ρ in km shown as the size of the points) and mean eDNA concentrations (in log copies/L) across species (colors) in panel B. Panel C indicating the signal amplitude (α^2) relation between non-spatial variance (σ^2) and species spatial autocorrelation (ρ in km shown as the size of the points).

Prediction accuracy on empirical data

Prediction error in empirical data decreased with increasing sampling density across all species (Appendix S1: Figure S1 and S2). However, this reduction was strongly modulated by the degree

of spatial autocorrelation (Appendix S1: Figure S2). Species characterized by short spatial autocorrelation (small ρ values) consistently exhibited higher prediction error (ε_1) across all sampling densities, whereas species with wider spatial autocorrelation structures (e.g., $\rho > 500$ km) showed substantially lower prediction errors (ε_1) even under comparatively sparse sampling regimes. Although increased sampling effort reduced error for all species (with varying degrees of spatial autocorrelation), patchily distributed taxa retained a persistently higher prediction error (ε_1).

Simulations

Prediction error in simulation data (ε_2) decreased non-linearly with increasing sampling effort and with higher spatial autocorrelation (Figure 4A). These patterns were supported by a log-log linear model where the prediction error (ε) decreased exponentially for every exponential increase in sampling density ($\beta = -0.176$) and spatial autocorrelation ($\theta = -0.228$), with an additional negative interaction ($\gamma = -0.0684$) and with an intercept of $\omega = 1.65$ (i.e., $\varepsilon = 5.2$ when E and $\rho = 1$). For instance, at a sampling intensity of 6 samples per 10,000 km², species with spatial autocorrelation values of 50 km had mean prediction errors of 1 (Figure 4A) indicating that on average every point predicted the eDNA concentration will differ from the predicted true concentration (Z) by a factor of 2.72 (for perspective a concentration of 100 copies/L will be predicted between 37-270 copies/L; Appendix S1: Figure S3). In contrast, for the same sampling effort (i.e., 6 samples per 10,000 km²) but species with spatial autocorrelation $\rho > 500$ km had errors below 0.4 indicating an average of over- and underestimation by a factor of 1.5 (for perspective a concentration of 100 copies/L will be predicted between 67-150 copies/L; Appendix S1: Figure S3). When quantifying the marginal reduction in prediction error ($\Delta\varepsilon_2$) across successive increases in sampling density (ΔE), improvements were greatest at low sampling effort and declined rapidly with additional

sampling, with $\Delta\epsilon_2$ approaching zero at higher sampling densities (Figure 4B). Marginal gains effectively plateaued when sampling effort increased above 10 samples per 10,000 km², depending on the ρ (Figure 4B). Importantly, diagnostic checks revealed no systematic bias in the predictions, so the observed variation reflects pure precision, (Appendix S1: Figure S4). For interpretative purposes, we adopt an arbitrary threshold of $\Delta\epsilon_2/\Delta E = -0.05$ to denote negligible marginal gains. However, this value is arbitrary and may be redefined according to study-specific accuracy requirements.

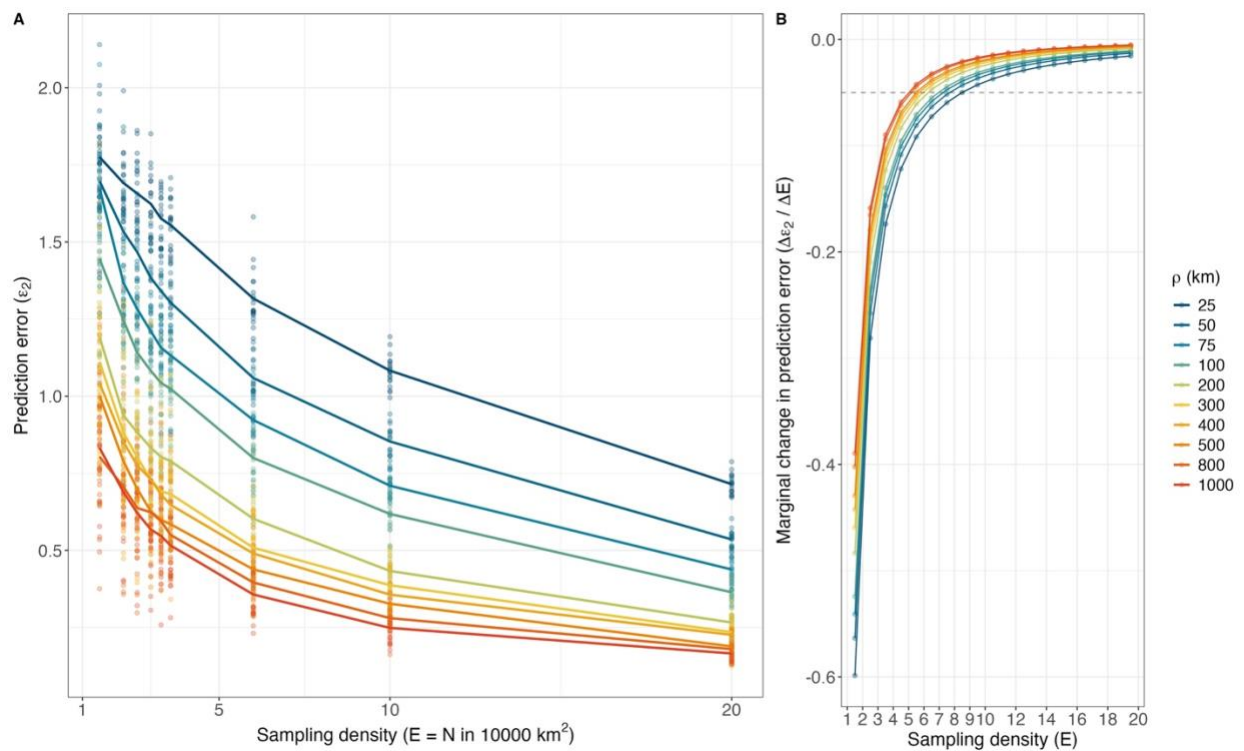


Figure 4. Estimated prediction error (ϵ_2) across sampling-effort scenarios (E ; indicated by sampling density in 100x100 km grid) for simulated datasets (A) across various spatial autocorrelation values (indicated in colors). Points represent individual simulation outcomes, while solid lines denote the corresponding mean trends across replicates. Marginal improvement (B) in prediction performance (quantified as the change in error - $\Delta\epsilon_2$) across successive increases in sampling densities ($\Delta E = 1$) where each point represents the marginal change at the midpoint of a unit interval (e.g., $E = 1.5$ reflects the

change from $E = 1$ to $E = 2$), with lines connecting successive points. Due to the discrete intervals of sampling effort in simulations, the lines in panel B are predicted from the fitted model in Eq. 6 (see Appendix S1: Figure S5 for model fit diagnostics). The dashed grey horizontal line indicates the arbitrary threshold ($\Delta\epsilon_2 / \Delta E = -0.05$) at which additional sampling effort yields negligible improvements in prediction accuracy.

GP parameter recovery

Sampling effort and spatial autocorrelation strongly influenced the ability of the GP model to recover the parameters used to generate the simulated spatial fields (Figure 6). Estimation error for the depth-specific mean (μ_d) decreased with increasing sampling density for simulated fields with small spatial autocorrelation but showed the opposite pattern for fields with large spatial ranges (Figure 5A). Specifically, τ (root mean squared error between estimated and simulated) for μ_d increased with sampling effort when spatial autocorrelation exceeded 400 km, indicating that mean estimates became increasingly biased in highly spatially structured fields. Additionally, the recovery of the spatial autocorrelation length-scale (ρ) was highly sensitive to sampling density (Figure 5B). At low sampling effort ($E < 4$), estimates of ρ showed substantial error, particularly for simulations with large spatial ranges. However, estimation accuracy improved rapidly as sampling effort increased, with τ declining sharply and stabilizing once sampling densities were higher than 6 samples per 10,000 km², beyond which additional sampling produced only marginal improvements in ρ estimation.

Estimation error for the amplitude parameter (α^2) declined with increasing sampling density for simulations with small spatial autocorrelation, while remaining relatively stable across sampling densities for simulations with larger spatial ranges (Figure 5C). Overall improvements in α^2

estimation were therefore limited once sampling effort increased above 6 samples per 10,000 km². The non-spatial variance parameter (σ^2) was consistently well recovered across sampling densities (Figure 5D). τ declined sharply with increasing sampling density and approached near-zero values once sampling densities exceeded 6 samples per 10,000 km².

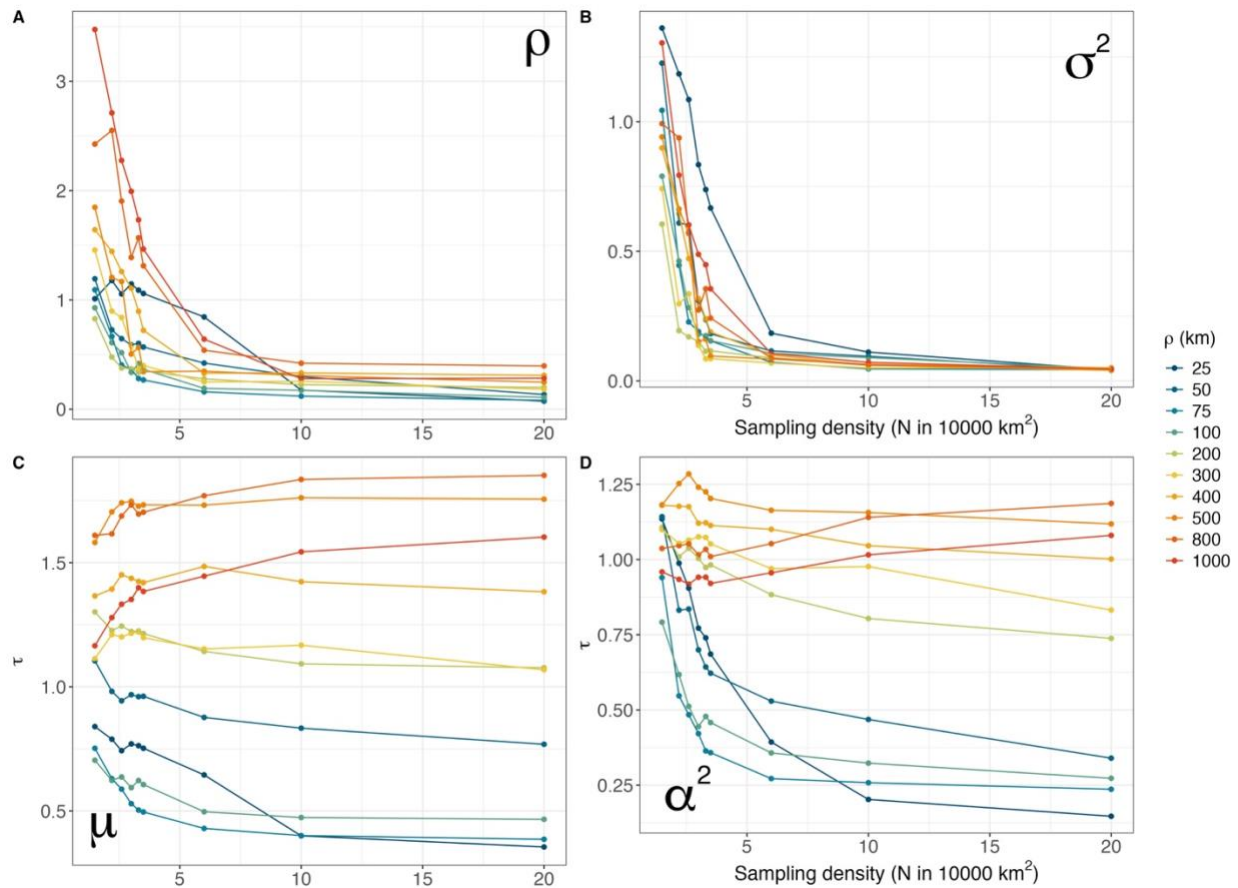


Figure 5. Accuracy of Gaussian Process (GP) parameter estimation across sampling effort (E ; expressed in sampling density) and spatial autocorrelation values (ρ) from simulation data. Each dot represents the mean of root mean squared error (τ) across 20 simulation iterations of the difference between estimated and true parameter values for (A) spatial autocorrelation (ρ), (B) non-spatial variance (σ^2), (C) depth-specific mean (μ_d), and (D) amplitude (α^2). Lines represent the regression of mean τ across sampling effort for each spatial autocorrelation level.

Comparison of prediction accuracy between empirical and simulation data

Prediction error (ε) followed similar patterns using both empirical and simulation datasets. Specifically, the estimated intercept (ω), the influence of sampling effort (β), the effect of spatial autocorrelation (θ), and their interaction term (γ) all showed consistent directional effects and aligned closely with the 1:1 line, demonstrating a high degree of structural agreement between the empirical and simulated error models (Appendix S1: Figure S1).

Discussion

We provide quantitative evidence for how species-specific spatial autocorrelation governs the efficiency of eDNA spatial surveys for spatial distribution predictions. We show that optimal sampling effort is not universal but depends critically on both the spatial autocorrelation of target species and the specific inferential objectives of the survey.

Diminishing returns of sampling effort on prediction accuracy

The value of additional sampling effort is not fixed but depends on the spatial structure of the target species (Xu et al., 2015; Zhao et al., 2022). Prediction error decreased non-linearly with sampling effort and with increasing spatial autocorrelation, confirming that homogeneously distributed species are inherently easier to reconstruct than patchier ones at equivalent effort (Wagner & Fortin, 2005). This interaction matters because it shows the value of additional samples is not universal, where the same increase in effort can produce substantial gains for one species and only marginal gains for another (García-Machado et al., 2024). Our results move beyond a general statement of sampling efficiency, providing a quantitative framework for understanding when additional effort is informative and when it yields diminishing returns.

426

427 Crucially, the decision of the sampling effort based on prediction accuracy (ε_2) cannot be reduced
428 to a single universal number, as it depends heavily on both the acceptable margin of error and the
429 spatial autocorrelation of the target organism. Our adopted threshold for negligible marginal gains
430 ($\Delta\varepsilon_2 = -0.05$) serves as a practical heuristic, but acceptable error is inherently context dependent.
431 For instance, studies requiring high-resolution quantitative estimates might justify high sampling
432 effort to achieve stricter precision, while, broad-scale biodiversity screenings might tolerate looser
433 precision, allowing for even sparser sampling (Perkins et al., 2021; Radinger et al., 2019)
434 Moreover, the species spatial autocorrelation parameter plays an additional role in defining the
435 decision on sampling effort. Highly autocorrelated species (e.g., $\rho > 500$ km) reach the precision
436 threshold of negligible improvement at much lower sampling densities (roughly 4 to 5 samples per
437 10,000 km²), conversely, patchily distributed species ($\rho < 50$ km) require a denser sampling
438 network (roughly 8 to 11 samples per 10,000 km²) before the marginal gains drop below the
439 threshold. Once the required precision and ρ are known (from prior literature, pilot sampling or
440 expert judgement), the log-log model $\varepsilon = e^{\omega} E^{\beta} \rho^{\theta} (E\rho)^{\gamma}$ (from Eq. 6; Figure 4B, Appendix S1: Figure
441 S1 and S6) can be inverted to solve for the minimum sampling density E required to keep $\Delta\varepsilon$ below
442 the chosen threshold.

443

444 Applying our simulation-derived thresholds to our empirical data further illustrates this context
445 dependency. Extrapolating the findings from the simulations, species (with high spatial
446 autocorrelations) such as *Leuroglossus stilbius* and *Microstomus pacificus* would require only 4
447 samples per 10,000 km² to achieve a negligible marginal gain in precision (defined here arbitrary
448 as $\Delta\varepsilon = -0.05$). In contrast such as *Trachurus symmetricus*, *Thaleichthys pacificus*, *Stenobranchius*

leucopsarus, and *Tarletonbeania crenularis* (species with low spatial autocorrelation) would require approximately 9 samples per 10,000 km² to reach the same relative threshold (Liu et al., 2026). The remaining species fall within the range of 4-9 samples per 10,000 km² (for context this is slightly higher than our current empirical grid and CalCOFI's standard 75-station grid with 3.6 and 3.4 samples per 10000 km² respectively), consistent with their intermediate spatial autocorrelation values. However, given that our analysis is based on 12 fish species, extending this conclusion to other taxa or ecosystems should be done with caution, and additional species may exhibit different scaling relationships.

Although we report effort per 10,000 km², the underlying point is simpler: what matters is how far apart samples are relative to a species' autocorrelation. A sampling effort E puts samples roughly $\sqrt{(10,000/E)}$ km apart, so the thresholds above can equivalently be stated as a ratio of sample spacing to length-scale: $d/\rho \approx 0.1$ for smoothly distributed species ($E \approx 4-5$ with $\rho \approx 500$ km), but $d/\rho \approx 0.6-0.7$ for patchily distributed ones ($E \approx 8-11$ with $\rho \approx 50$ km). The same reasoning should therefore carry to other systems and scales, though the ratio is not a single fixed number, because error depends on E and ρ jointly (Eq. 6).

Depth-dependent patterns

Surface-associated species were consistently patchier than deep-associated ones and showed larger surface–deep differences in mean concentration (Figure 3), which is consistent with greater spatial heterogeneity in surface waters driven by stronger horizontal gradients, mixing, productivity hotspots, and advective transport, while deeper layers may be more spatially structured and comparatively homogeneous at the regional scale (Guri et al., 2023). From a sampling-efficiency

perspective, this implies that surface-focused monitoring may require denser spatial sampling to reconstruct patchy distributions, whereas deep-layer monitoring may allow coarser sampling grids while retaining predictive fidelity-provided that the eDNA field at depth remains spatially coherent and that vertical transport does not dominate the observed signal.

Estimating GP parameters from reduced-effort simulations

Estimation of μ_d improved with increasing effort when scenarios had small spatial autocorrelations, but for highly autocorrelated scenarios (especially above 400 km) error (τ) in estimating μ increased rather than declined. One likely explanation is that, when the spatial autocorrelation parameter approaches or exceeds the spatial extent of the study area (200 km longitudinal here), the data provide only weak information about the parameter, leaving the posterior estimates heavily dependent on the Bayesian prior.

By contrast, estimation of the spatial autocorrelation parameter ρ was substantially more sensitive to reduced sampling effort, particularly in scenarios with low spatial autocorrelation. Estimating the mean mainly requires observations that broadly represent the range of values across the study area, whereas estimating spatial autocorrelation requires information on how similarity between samples changes with distance (Paciorek, 2010; Rocha et al., 2021). In practical terms, μ_d can remain relatively robust under reduced sampling effort, provided that the spatial structure is still contained within the extent of the survey. By contrast, estimating ρ depends much more strongly on the distances among sampling points, because resolving spatial autocorrelation requires enough closely and broadly spaced pairs to detect how similarity declines with distance (Paciorek, 2010).

Overall, this behavior is expected when the underlying process varies at a finer spatial scale than the sampling design can resolve. Once the true ρ is smaller than the spatial resolution of the sampling grid (e.g., a grid spaced at 5 km cannot resolve patchiness occurring at a 1 km scale) the model has no way to detect the pattern, and predictions default to the priors with large residual uncertainty (Kaufman & Shaby, 2013). Under these conditions, the model is operating close to the limit of what can be learned from the sampling design, and additional changes in true ρ may not translate into detectable changes in map-level accuracy. Practically, this implies that sample-removal diagnostics (or any assessment that treats an undersampled reference as truth) can become uninformative for very patchy species, because the benchmark surface itself is not resolved well enough to support fine-scale comparisons (Kaufman & Shaby, 2013). This, in turn, motivates our reliance on simulations with much higher spatial resolution to characterize how sampling effort should scale with spatial autocorrelation.

Overall, the parameter estimates indicate that most gains in estimation of the GP parameters occurred at low sampling effort, with little additional improvement once effort exceeded roughly 6 samples per 10,000 km² per depth layer. This consistent leveling-off across GP parameters estimation (Figure 5) suggests a clear point of diminishing returns, beyond which further increases in sampling effort are unlikely to substantially improve inference under the spatial scales considered here.

Limitations on extrapolating simulations on real-world scenarios

The simulated and empirical reduced effort analysis showed strong agreement. This concordance suggests that our simulation framework captured important features of eDNA spatial structure at

the scale of ten-degree latitude. At the same time, extrapolating directly from simulations to natural systems requires care because real eDNA fields may violate the simplifying assumptions of the stationary, isotropic squared exponential kernel. Spatial structure may vary across the domain (non-stationarity), differ by direction (anisotropy), or include sharp transitions (fronts, plume boundaries, shelf-break structure) that are not well represented by a globally smooth covariance function (Paciorek & Schervish, 2006). In addition, eDNA observation error is unlikely to be purely Gaussian, with variability arising from sources such as PCR bias, compositional constraints, incomplete calibration, and heterogeneity in filtration or extraction efficiency (Bessey et al., 2020; Gloor et al., 2017; Kelly et al., 2019; Shaffer et al., 2025). Our framework accounts for these sources (Guri et al., 2025), so they are unlikely to bias the results presented here. Finally, ecological processes such as migration, schooling, habitat discontinuities, or patchy prey fields can generate multi-scale patterns that depart from a single characteristic length-scale (Wagner & Fortin, 2005). These features may help explain elevated prediction error at intermediate autocorrelation levels in the empirical data relative to simulations, where additional sources of spatial complexity are not represented. Despite these real-world complexities, the strong concordance between empirical and simulated results indicates that our framework captures a robust and ecologically meaningful relationship between spatial autocorrelation and the sampling effort required to estimate it reliably.

Furthermore, the simulation assumes a stationary GP with fixed amplitude and noise and a single length-scale across depths, whereas real eDNA fields may have variable variance structure, multiple characteristic scales, and dependence on environmental covariates such as temperature, salinity, chlorophyll, and bathymetry (Liu et al., 2026). Our prediction error (ϵ_2) is also computed

against a "complete" prediction from a 4,000-point subset (Z) rather than the full grid, a limitation most visible in the secondary thinning comparison, where the reference field is constrained to N = 358 samples, making prediction error higher and more stochastic.

Intrinsically, eDNA integrates ecological information through space and across time via transport and decay (Brandão-Dias et al., 2023; Fukaya et al., 2021). This means that even if organisms are locally patchy, the observed eDNA field may be smoother than the organism field, depending on hydrodynamic conditions and degradation rates. In our framework, such integration is embedded within the inferred length-scale, reinforcing that ρ should be interpreted as an effective length scale of the observed eDNA field rather than a purely ecological niche parameter (Jo et al., 2025). Importantly, because the length-scale parameter here reflects the integrated effects of organism distribution together with eDNA transport and decay, the depth patterns observed in this study may also reflect depth-dependent physical processes (Brandão-Dias et al., 2023). Integrating eDNA observations with independent abundance data (trawl, acoustics) would allow some degree of separation of the spatial autocorrelation parameter, enabling clearer ecological interpretation of ρ and improving the transferability of sampling-effort recommendations (Guri et al., 2024).

Spatial scale also sets an important boundary on the applicability of these results. The relationships identified here are expected to hold most strongly when interpreted relative to the extent and resolution of the study domain, rather than as fixed thresholds transferable across all systems. In a small study domain, the autocorrelation scale of the observed eDNA field may exceed the survey extent, causing the field to appear nearly uniform and making μ_d difficult to estimate independently of the spatial process (Paciorek, 2010; Redfern et al., 2006).

Implications for survey design and management applications

Sampling effort required for eDNA surveys should be matched to the intended inferential goal. When the objective is to estimate GP parameters such as the mean (μ_d) for deriving depth-specific or region-specific abundance indices typically used in stock assessment indices (Shelton et al., 2022), the autocorrelation (ρ) for applied ecological questions, like implementing protected area networks or monitoring programs (Laroche, 2024), or the variance parameter (α^2), random sampling designs that adequately cover the full spatial domain of the species spatial autocorrelation are likely to be sufficient (Rocha et al., 2021), as sampling effort appears to have relatively limited influence on parameter estimation (Figure 5). In contrast, when the objective shifts toward predicting spatial distributions - such as identifying hotspots, mapping species occurrence, or quantifying spatial niche breadth - the accuracy of these predictions is much more sensitive to both sampling effort and the underlying spatial autocorrelation of the species (Figure 4A). Prediction is the more demanding of the two because it requires both accurate covariance parameters (Eq. 1–2) and observed points close enough to the location being predicted (Eq. 3).

This distinction also clarifies how sampling design shapes inference. Estimation of the mean benefits primarily from coverage (sampling across the full domain) and secondarily from balance (avoiding over-sampling a limited region) (Rocha et al., 2021). By contrast, estimation of the length-scale parameter requires not only broad coverage but also a wide distribution of pairwise distances, including closely spaced stations that resolve rapid decorrelation and widely spaced stations that constrain the long-distance tail of the covariance function (Kaufman & Shaby, 2013).

Taken together, these results suggest that the framework translates into a small set of design considerations that can be resolved before a cruise is planned. The starting point is a clear statement of the inferential objective and the precision it demands, since this determines whether effort should prioritize spatial coverage or the resolution of fine-scale autocorrelation. This is complemented by a prior estimate of the target taxon's spatial autocorrelation, drawn from published geostatistical fits, a pilot survey, or the depth-related pattern reported here, which allows Eq. 6 to be used to characterize the relationship between sampling density (E) and expected accuracy. From this relationship, a sampling density can be identified beyond which additional effort yields diminishing returns relative to the precision required. Because the length-scale parameter cannot be recovered from a uniformly thinned grid, the resulting design should also be assessed for its inclusion of both widely and closely spaced station pairs. In multispecies programmes, these considerations are most appropriately governed by the patchiest taxon the survey is intended to resolve.

Applied to existing programs this yields a concrete conclusion. The West Coast eDNA grid analyzed here (3.6 samples per 10,000 km² per depth layer) and the CalCOFI 75-station grid (3.4 samples per 10,000 km²) already sit close to the effort at which returns diminish for smoothly distributed species and for estimating depth-specific means, so programs whose primary product is an abundance index need not expand. Programs intended to map patchily distributed taxa, including several forage and mesopelagic species examined here, are under-sampled by a factor of roughly two to three, and that shortfall is spatial rather than analytical (it cannot be recovered by sequencing more deeply or filtering more water at stations already occupied).

608 For managers and survey designers the practical message is that effort should be set by the
609 patchiness of the taxa a programme is required to resolve and by the decision the data must support,
610 and that both can be specified in advance rather than discovered afterwards. Nevertheless, across
611 the scenarios considered here and the spatial domain used in this study, sampling effort of 4 - 9
612 samples per 10,000 km² per depth layer marked the point at which diminishing returns became
613 apparent given species-specific spatial autocorrelation.

614

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Conflict of interest statement

The authors declare no conflict of interest.

Data availability statement

The data has been submitted to Zenodo repository with DOI number 10.5281/zenodo.21141367 and the code can be found on GitHub repository <https://github.com/gledguri/Smart-eDNA-sampling>.

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